

Online 1 (A1/A2): Tired of flights

Time: 30 mins

Problem description

Let me remind you the story of offline 1.

You are analyzing a global flight network centered around Dhaka (DAC).

A traveler wants to fly from Syreja International Terminal Hazrat Shahjalal International Airport to London Heathrow Airport. Each flight between countries has a fixed ticket price. The traveler has one international discount coupon that can be used to halve the cost of any one flight in the journey. (If the cost is x , it becomes $\lfloor x/2 \rfloor$)

You are given a list of international flights with their prices. Find the minimum possible travel cost from Dhaka to London.

Now, you are tired of flights and want to find the minimum possible travel cost from Dhaka to London within k flights. Of course, you have the coupon.

Required complexity

Time complexity: $O(VE)$

Input

The first input line contains three integers n , m , and k :

- n is the number of airports
- m is the number of international flight connections
- k is the maximum number of flights the traveler can take

The airports are numbered $1, 2, \dots, n$.

Airport 1 is **Hazrat Shahjalal International Airport (Dhaka)**, and airport n is **London Heathrow Airport**.

After this, there are m lines describing the flights.

Each line contains three integers a , b , and c , meaning that there is a **unidirectional flight** from airport a to airport b with ticket price c .

Output

Print a single integer:

the **minimum possible travel cost** from Dhaka to London using **at most k flights**, after optimally applying the discount coupon. If it is not possible to reach London within k flights, print "Not possible".

Example

Input	Output
4 5 1 1 2 4 2 4 4 1 3 10 3 4 1 2 3 2	Not possible
4 5 2 1 2 4 2 4 4 1 3 10 3 4 1 2 3 2	6
4 5 3 1 2 4 2 4 4 1 3 10 3 4 1 2 3 2	5